31 fixes

31 copy-paste prompts

app-performance toolkit.

Every fix you need to make your vibe-coded app *feel instant*, explained simply, with a ready-to-use AI prompt for each one.



App-performance toolkit

Checklist | Prompts

- [x] Skeleton screens
- [x] WebP or AVIF images
- [x] No duplicate API calls
- [x] Debounced inputs
- [ ] + 27 more

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BEFORE YOU START

How to use this *toolkit.*

This toolkit covers all 31 fixes from the video, each with a diagram that shows the idea at a glance. Each one gets a card that explains what it is, why it matters, how to do it, what to watch out for, and how to check it worked, plus a prompt you can paste straight into your AI coding tool.

01

Measure first

Run the baseline prompt so you know your starting point. Without a before number, you can't tell what helped.

02

Run the audit

Use the Start Here prompt to let your AI tool check your app against all 31 items and rank what matters for *your* app.

03

Fix one item at a time

Commit your code, paste one prompt, review the plan, then let it build. Small changes are easier to test and undo.

04

Verify and repeat

Use each card's verify step, re-measure, and keep going down the list. Finish with the final check prompt.

Anatomy of a card

Number & title
Matches the order in the video.

Category
Where in your app the fix lives.

Impact
How much users will notice it.

Effort
Roughly how much work it takes.

Frontend · Assets · Network · Backend · Database · AI · Infra · Monitoring

Prompt tips

- Ask for a plan first: add **“Show me your plan before changing any code.”** to the end of any prompt.
- Not every item applies to every app. Skip anything the audit marks as not relevant.
- Tell your AI tool your stack if it can't see your code (framework, host, database).
- Run prompts in a fresh chat or session when your context gets long, so the AI stays focused.

PROMPT 00

Start *here.*

Run these two prompts before anything else. The first records where you are today. The second checks your app against all 31 items and tells you which ones to do first.

Step 1 · Measure your baseline

BASELINE PROMPT

```
Before we optimize anything, help me measure my app's current speed so we can compare later.

1. Pick my 5 most important pages and 5 most-used API endpoints.
2. For each page, tell me how to record: LCP, INP, CLS, total JS size, total image weight, and number of requests (mobile audit with throttling).
3. For each endpoint, measure the average and slowest response times and the number of database queries per request.
4. Save the results in a PERFORMANCE.md file in my repo as a "Before" table.

After each fix, we'll add an "After" column.
```

Step 2 · Run the full audit

AUDIT PROMPT

```
Act as a senior performance engineer. Audit my entire app for speed problems before changing anything.

1. Detect my stack: framework, hosting, database, ORM, and AI provider (if any).
2. Check my app against this 31-point checklist: skeleton screens, image formats, duplicate API calls, debouncing, pagination, CDN, background jobs, N+1 queries, compression, caching, unused dependencies, external API timeouts, AI streaming, optimistic UI, lazy-loading, parallel requests, database indexes, tracing, server/database region, server/static rendering, third-party scripts, code splitting, list virtualization, selected columns, connection pooling, re-renders, browser caching, load balancing, fonts, real-user monitoring, and slow-device testing.
3. For each item, mark it DONE, NEEDS WORK, or NOT RELEVANT, with one line explaining why.
4. Give me a prioritized top-10 fix list, ordered by impact on real users versus effort.

Don't change any code yet. Just give me the report.
```

> **Keep the report**
> Save the audit's top-10 list somewhere you'll see it. Then work through it one item at a time, using the matching card and prompt from this toolkit.

THE CHECKLIST

All 31 fixes *at a glance.*

	#	FIX	CATEGORY	IMPACT	EFFORT
☐	01	Add skeleton loading screens	Frontend	Medium	Low
☐	02	Serve images in WebP or AVIF	Assets	High	Low
☐	03	Prevent duplicate API calls	Network	High	Medium
☐	04	Debounce search and autocomplete inputs	Network	Medium	Low
☐	05	Paginate API results	Backend	High	Medium
☐	06	Serve static files through a CDN	Infra	High	Low
☐	07	Move slow tasks to a background job queue	Backend	High	Medium
☐	08	Fix N+1 queries	Database	High	Medium
☐	09	Enable Brotli or Gzip compression	Network	Medium	Low
☐	10	Cache frequently accessed data with Redis	Backend	High	Medium
☐	11	Remove unused dependencies	Assets	Medium	Low
☐	12	Set timeouts for external API calls	Backend	High	Low
☐	13	Stream AI responses token by token	AI	High	Medium
☐	14	Update the UI before the server responds	Frontend	Medium	Medium
☐	15	Lazy-load images below the screen	Assets	Medium	Low
☐	16	Run independent requests in parallel	Backend	High	Low
☐	17	Add database indexes for common queries	Database	High	Low
☐	18	Trace slow backend requests with OpenTelemetry	Monitoring	High	Medium
☐	19	Place servers close to the database	Infra	High	Low
☐	20	Render pages on the server or at build time	Frontend	High	Medium
☐	21	Load third-party scripts after the page	Frontend	High	Low
☐	22	Split JavaScript by page	Frontend	High	Medium
☐	23	Virtualize long lists	Frontend	Medium	Medium
☐	24	Select only the columns you need	Database	Medium	Low
☐	25	Use database connection pooling	Database	High	Low
☐	26	Prevent unnecessary re-renders	Frontend	Medium	Medium
☐	27	Enable browser caching for static files	Network	Medium	Low
☐	28	Load-balance traffic across multiple app instances	Infra	Medium	Medium
☐	29	Self-host fonts and limit font weights	Assets	Medium	Low
☐	30	Track real-user speed with a performance monitoring tool	Monitoring	High	Low
☐	31	Test on slow phones and networks	Monitoring	High	Low

SHORT ON TIME?

Quick *wins.*

These fixes are **high impact** and **low effort**. If you only have an afternoon, start with these.

- 02 Serve images in WebP or AVIF
- 06 Serve static files through a CDN
- 12 Set timeouts for external API calls
- 16 Run independent requests in parallel
- 17 Add database indexes for common queries
- 19 Place servers close to the database
- 21 Load third-party scripts after the page
- 25 Use database connection pooling
- 30 Track real-user speed with a performance monitoring tool
- 31 Test on slow phones and networks

A note on order

The cards follow the order of the video, not order of importance. The audit prompt on the Start Here page ranks them for your specific app, so use its top-10 list as your real to-do list.



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for more guides on building fast, secure, and cost-efficient apps.

01
/ 31

Add skeleton loading screens.

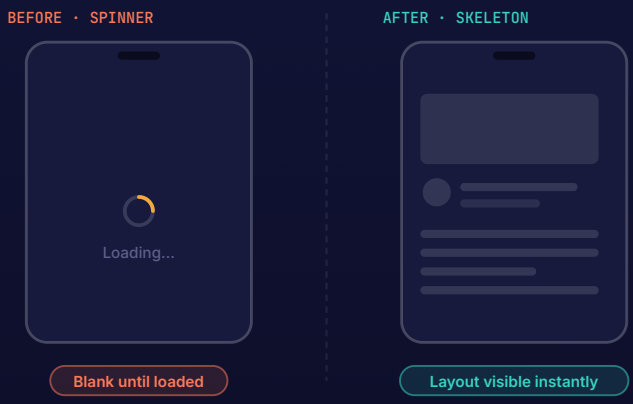
Frontend IMPACT Medium EFFORT Low

WHAT IT IS

Show grey placeholder shapes in the exact layout of the content that's about to load, instead of a blank screen or a lone spinner.

WHY IT MATTERS

Users perceive the app as faster because they can see structure immediately, and the page doesn't jump when the real content arrives.



A skeleton shows the page's shape while data loads.

HOW TO DO IT

- Build skeletons that match the final layout's size and position (cards, rows, avatars, text lines).
- Show them only while data is loading, and swap them out the moment it arrives.
- Add `aria-busy="true"` to the loading region so screen readers know it's loading.

WATCH OUT

A skeleton with different dimensions than the real content causes layout shift (CLS). Disable the shimmer animation for users with `prefers-reduced-motion`.

HOW TO VERIFY

Throttle the network in DevTools and reload: you should see the layout instantly, with no jumping when data appears.

COPY-PASTE PROMPT

```
Add skeleton loading screens to my app.

1. Find every screen and component that shows a blank area, a spinner, or "Loading..." while fetching data.
2. For each one, build a skeleton that matches the final layout (same widths, heights, spacing, and number of rows/cards).
3. Use one reusable Skeleton component that matches my existing styling.
4. Add aria-busy="true" on loading regions and turn off the shimmer animation when prefers-reduced-motion is set.
5. Make sure there is no layout shift when the real content replaces the skeleton.

Don't change any data-fetching logic. When done, list every screen you updated.
```

02 / 31

Serve images in WebP or AVIF.

Assets · IMPACT High · EFFORT Low

WHAT IT IS

Convert JPEG and PNG images to modern formats, and resize them to the size they're actually displayed at.

WHY IT MATTERS

Images are usually the heaviest thing on a page. WebP and AVIF files are typically much smaller than JPEG or PNG at similar quality.



HOW TO DO IT

- Use your framework's image component if it has one; it usually converts, resizes, and lazy-loads for you.
- Otherwise use `<picture>` with an AVIF source, a WebP source, and a JPEG/PNG fallback.
- Serve multiple sizes with `srcset` and `sizes` so phones don't download desktop-sized images.

WATCH OUT

Never ship a 4000px photo into a 400px slot. Always set `width` and `height` (or an aspect ratio) to prevent layout shift.

HOW TO VERIFY

In DevTools → Network → Img, check that images show type avif or webp and that their file sizes dropped.

COPY-PASTE PROMPT

```
Optimize all images in my app.

1. Find every image: <img> tags, CSS backgrounds, and images in components.
2. If my framework has a built-in image component, use it. Otherwise serve AVIF and WebP via
<picture> with a JPEG/PNG fallback.
3. Resize images to the sizes they're displayed at and add responsive srcset/sizes.
4. Compress them (quality around 75-80 is usually visually lossless).
5. Set width and height (or aspect-ratio) on every image to prevent layout shift.
6. Do NOT lazy-load the main image at the top of the page; give it high fetch priority
instead.

Show me a before/after table of total image weight per page.
```

03
/ 31

Prevent duplicate API calls.

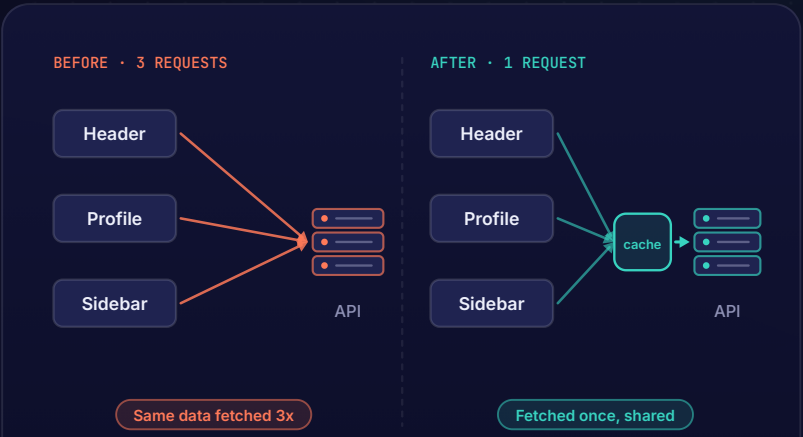
Network IMPACT High EFFORT Medium

WHAT IT IS

Stop the same request from being sent several times at once, for example by multiple components, double-clicks, or re-renders.

WHY IT MATTERS

Duplicate calls waste bandwidth, slow everything down, overload your server, and can even create duplicate records or charges.



Components share one request instead of each fetching.

HOW TO DO IT

- Share one in-flight request between components that need the same data, and cache the result.
- A data-fetching library with built-in caching and deduplication handles most of this automatically.
- Disable submit buttons while a request is pending, and use idempotency keys for create/payment endpoints.

WATCH OUT

In React development mode, StrictMode intentionally runs effects twice. That's normal in dev, but make sure it doesn't happen in production.

HOW TO VERIFY

Open DevTools → Network, load each page, and confirm each endpoint is called once, not several times.

COPY-PASTE PROMPT

```
Find and eliminate duplicate API calls in my app.

1. Load each page and trace which endpoints get called more than once for the same data.
2. Find the cause: multiple components fetching the same thing, effects re-running, missing dependency arrays, or no caching.
3. Fix it by sharing data (a cached data-fetching layer or shared state) so each piece of data is fetched once and reused.
4. Disable submit buttons while requests are pending so double-clicks can't send twice.
5. For create, payment, or other write endpoints, add idempotency keys so a repeated request isn't processed twice.

Ignore the double-run caused by React StrictMode in development only. List each duplicate you found and how you fixed it.
```

04 / 31

Debounce search and autocomplete inputs.

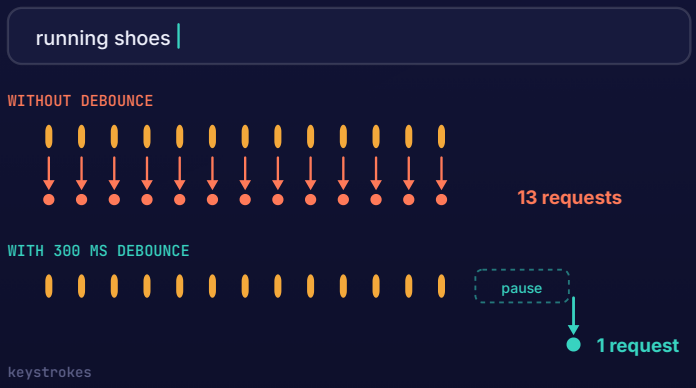
Network | IMPACT Medium | EFFORT Low

WHAT IT IS

Wait until the user pauses typing (around 250–400 ms) before sending a search request, instead of firing one on every keystroke.

WHY IT MATTERS

Typing "running shoes" without debouncing sends 13 requests. With debouncing it sends one or two.



One request after the user pauses, not one per key.

HOW TO DO IT

- Wrap the search call in a debounce of about 300 ms.
- Cancel the previous request when a new one starts (for example with `AbortController`).
- Skip searches for very short queries (for example, fewer than 2 characters).

WATCH OUT

Without cancelling, an older slow response can arrive after a newer one and overwrite the correct results.

HOW TO VERIFY

Type quickly in the search box with the Network tab open: you should see one request after you stop typing.

COPY-PASTE PROMPT

```
Debounce every search, filter, and autocomplete input in my app.

1. Find all inputs that trigger an API call or expensive filtering as the user types.
2. Add a debounce of about 300 ms using one reusable hook/utility.
3. Cancel the previous in-flight request when a new one starts (AbortController), so
stale results never overwrite new ones.
4. Don't search until the query has at least 2 characters, and show a small loading
indicator while searching.
5. Keep the input itself instant; only the request should be delayed.

List every input you changed.
```

05
/ 31

Paginate API results.

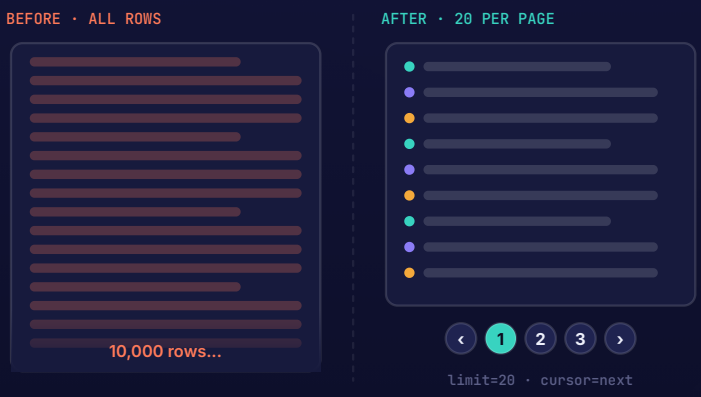
Backend IMPACT High EFFORT Medium

WHAT IT IS

Return data in pages (for example 20–50 items at a time) instead of sending every row in one response.

WHY IT MATTERS

Endpoints that return everything get slower as your data grows, and eventually time out or crash the browser.



Send a page of results, not the whole table.

HOW TO DO IT

- Add a limit and a cursor (or page) parameter to every list endpoint.
- Prefer cursor-based (keyset) pagination for large or fast-changing tables; it stays fast where offset pagination slows down.
- Enforce a maximum page size on the server, so nobody can request a million rows.

WATCH OUT

Offset pagination (`OFFSET 100000`) gets slower the deeper you go, and can skip or repeat rows when data changes between pages.

HOW TO VERIFY

Call each list endpoint and confirm it returns a limited number of items plus a next-page cursor.

COPY-PASTE PROMPT

```
Add pagination to every list endpoint in my app.

1. Find all API routes and queries that return lists without a limit.
2. Add cursor-based pagination (limit + cursor, ordered by a unique, indexed column) and return a nextCursor.
3. Default to 20 items per page and enforce a maximum (e.g., 100) on the server.
4. Make sure the column used for ordering/cursoring is indexed.
5. Update the frontend to use "Load more", infinite scroll, or page buttons, whichever fits each screen.

Don't break existing API consumers; tell me if any response shape had to change.
```

06
/ 31

Serve static files through a CDN.

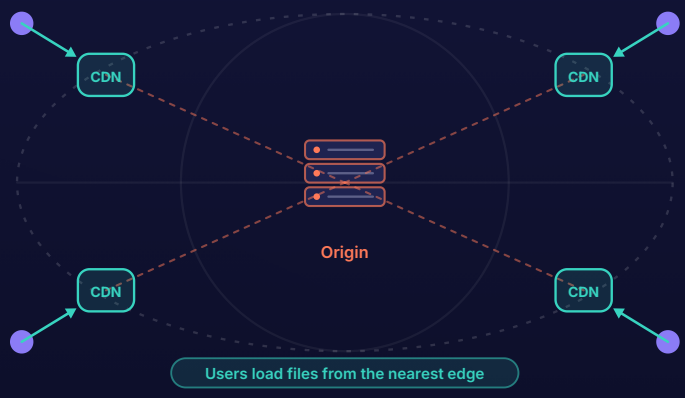
Infra IMPACT High EFFORT Low

WHAT IT IS

Deliver images, JavaScript, CSS, and fonts from a content delivery network that caches copies on servers around the world.

WHY IT MATTERS

Files load from a location near each user instead of travelling all the way to your origin server.



Static files are served from an edge near each user.

HOW TO DO IT

- Check your hosting platform first; many already put static files behind a CDN.
- Otherwise put a CDN in front of your static assets or storage bucket.
- Use versioned (hashed) file names so you can cache them for a long time.

WATCH OUT

Never let the CDN cache personalized or logged-in responses, or one user may see another user's page.

HOW TO VERIFY

Check the response headers of a static file for a cache status such as HIT, and compare load times from different regions.

COPY-PASTE PROMPT

```
Make sure all my static assets are served through a CDN.

1. Check how my app is hosted and whether static files (JS, CSS, images, fonts, videos) already go through a CDN.
2. If not, set it up for my stack and point static asset URLs at it.
3. Make sure asset file names are versioned (content hashes) so they can be cached long-term.
4. Confirm that HTML, API responses, and anything personalized or behind login are NOT cached by the CDN.
5. Explain how to verify cache hits in the response headers.

Tell me about any manual setup I need to do in my hosting dashboard.
```

07
/ 31

Move slow tasks to a background job queue.

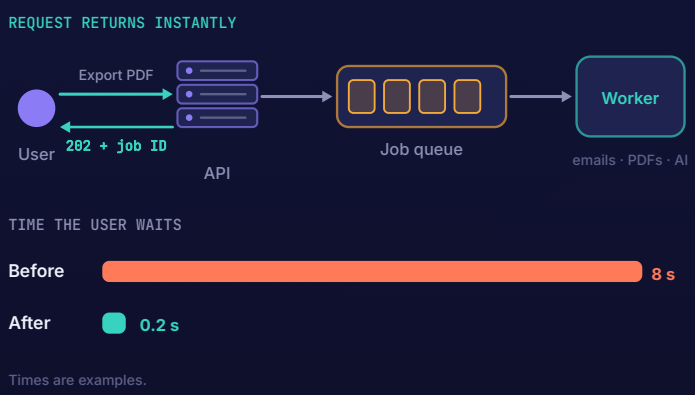
Backend IMPACT High EFFORT Medium

WHAT IT IS

Run slow work (sending emails, generating PDFs, processing videos, bulk AI calls) in a background worker instead of making the user wait for it.

WHY IT MATTERS

The request returns instantly, and heavy work no longer blocks your server from handling other users.



Slow work moves to a worker; the user isn't kept waiting.

HOW TO DO IT

- Push the task onto a job queue and immediately return a job ID to the user.
- Process jobs in a separate worker with retries, backoff, and a timeout.
- Show progress or notify the user when the job is done.

WATCH OUT

Make jobs safe to run twice (idempotent), because retries happen. On serverless platforms, work started after the response is sent may be killed, so use a real queue.

HOW TO VERIFY

Trigger the slow action: the request should return in well under a second, and the result should appear when the job finishes.

COPY-PASTE PROMPT

```
Move slow work out of my request handlers into a background job queue.

1. Find every API route that takes more than ~1 second because of work like sending emails,
generating files, image/video processing, webhooks, or bulk AI calls.
2. Recommend a job queue that fits my current stack and hosting, and set it up.
3. Change those routes to enqueue a job and immediately return a job ID (HTTP 202).
4. Process jobs in a worker with retries, exponential backoff, a max attempt count, and a
timeout.
5. Make each job idempotent so a retry never sends two emails or charges twice.
6. Add a way for the frontend to check job status or get notified when it's done.

List each task you moved and its old vs. new response time.
```

08
/ 31

Fix N+1 queries.

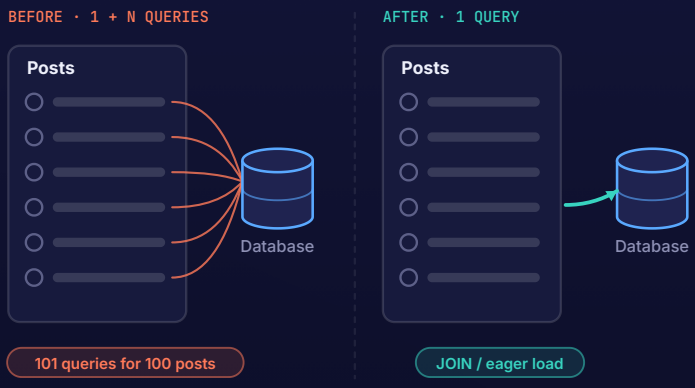
Database IMPACT High EFFORT Medium

WHAT IT IS

An N+1 problem is when your code runs one query to get a list, then one extra query for every item in that list.

WHY IT MATTERS

Loading 100 posts with their authors can turn into 101 database queries instead of 1 or 2, and it's one of the most common slowdowns in generated code.



Fetch related data in one query instead of one per row.

HOW TO DO IT

- Turn on query logging and count queries per request.
- Fetch related data in one go with a JOIN or your ORM's include/eager-loading option.
- In GraphQL or nested resolvers, batch lookups so each type is loaded once per request.

WATCH OUT

Eager-loading everything can over-fetch. Only include the relations the page actually uses.

HOW TO VERIFY

With query logging on, load the page: the number of queries should stay the same whether there are 10 items or 1,000.

COPY-PASTE PROMPT

```
Find and fix N+1 queries in my app.

1. Enable query logging and identify requests where the number of queries grows with the number of items returned.
2. Look especially for database calls inside loops, .map() calls, nested components, or GraphQL resolvers.
3. Fix each one by fetching related data in one query (JOIN or my ORM's include/eager-loading) or by batching lookups.
4. Only load the relations each page actually needs.
5. Keep the response data exactly the same.

For each fix, show the query count before and after.
```

09 / 31

Enable Brotli or Gzip compression.

Network · IMPACT Medium · EFFORT Low

WHAT IT IS

Compress text responses (HTML, CSS, JavaScript, JSON, SVG) before sending them, so less data travels over the network.

WHY IT MATTERS

Text files often shrink by 70% or more, which is a big win on slow mobile connections.



Text is compressed on the way out; media is skipped.

HOW TO DO IT

- Check your host or CDN first; many enable compression by default.
- Prefer Brotli and fall back to Gzip for older clients.
- For static files, pre-compress them at build time so the server doesn't compress on every request.

WATCH OUT

Don't compress files that are already compressed, like JPEG, PNG, WebP, AVIF, MP4, or ZIP. It wastes CPU and saves nothing.

HOW TO VERIFY

In DevTools → Network, check that text responses have a `content-encoding` header of br or gzip.

COPY-PASTE PROMPT

```
Make sure compression is enabled for all text responses in my app.

1. Check whether my host, CDN, or server already applies Brotli or Gzip compression.
2. If not, enable it for HTML, CSS, JS, JSON, SVG, and other text responses (Brotli preferred, Gzip fallback).
3. Pre-compress static assets at build time if my setup supports it.
4. Exclude already-compressed formats (images, video, archives) and tiny responses.
5. Make sure streaming responses (like AI output) still stream and aren't buffered by compression.

Show me how to confirm it with the content-encoding header.
```

10
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Cache frequently accessed data with Redis.

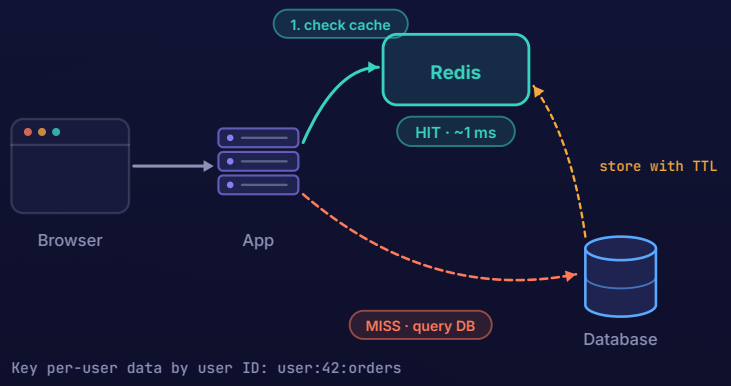
Backend IMPACT High EFFORT Medium

WHAT IT IS

Keep the results of frequent, expensive queries in Redis (or a similar in-memory store), so most requests skip the database entirely.

WHY IT MATTERS

Reading from memory takes about a millisecond, while a slow database query can take hundreds.



Most reads hit Redis; only misses reach the database.

HOW TO DO IT

- Use the cache-aside pattern: check the cache, fall back to the database on a miss, then store the result.
- Give every cache entry an expiry time (TTL), and delete or update it when the underlying data changes.
- Start with data that's read often and changes rarely: settings, product lists, public profiles, counts.

WATCH OUT

If a cache key doesn't include the user or account ID, one user can see another user's data. Never cache user-specific data under a shared key.

HOW TO VERIFY

Log cache hits and misses: repeat requests should be hits, and response times should drop.

COPY-PASTE PROMPT

```
Add Redis caching for my most frequently accessed data.

1. Find the most frequent and slowest read queries (use logs or tracing if available).
2. Pick the best candidates: data that's read often and changes rarely.
3. Implement cache-aside with a small reusable helper: get from cache → on miss, query the DB → store with a TTL.
4. Invalidate or update the cache whenever that data is written.
5. IMPORTANT: for any user- or account-specific data, include the user/account ID in the cache key, so one user can never see another user's data.
6. If Redis is unavailable, fall back to the database instead of crashing.

List what you cached, the key format, and the TTL for each.
```

11
/ 31

Remove unused dependencies.

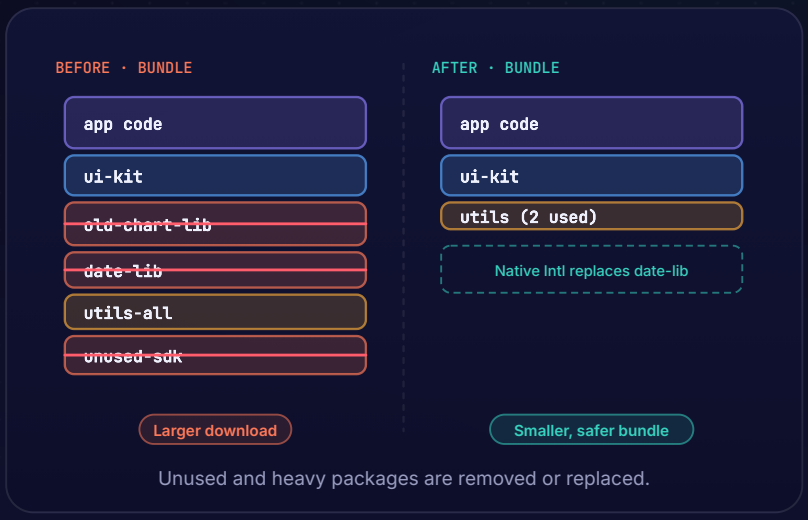
Assets IMPACT Medium EFFORT Low

WHAT IT IS

Delete packages you no longer use, and replace heavy libraries with lighter alternatives or built-in browser features.

WHY IT MATTERS

AI tools often install packages and never remove them. Unused code bloats your bundle, slows installs and builds, and adds security risk.



Unused and heavy packages are removed or replaced.

HOW TO DO IT

- Run a dependency-checking tool to list unused packages, and a bundle analyzer to find the heaviest ones.
- Replace heavy libraries where native features work (for example, date formatting with `Intl`).
- Import only what you use instead of whole libraries.

WATCH OUT

Some packages are used only in config files, scripts, or at build time. Double-check before deleting, and run a full build and tests afterwards.

HOW TO VERIFY

Compare bundle size and install size before and after, and make sure the app builds and runs normally.

COPY-PASTE PROMPT

```
Clean up the dependencies in my project.

1. List every package in my package file and check whether it's actually imported or used anywhere (including config files and scripts).
2. Remove the ones that are truly unused.
3. Analyze the production bundle and list the 10 heaviest dependencies.
4. For each heavy one, suggest a lighter alternative or a native browser/runtime feature, and swap it only if it's safe.
5. Change whole-library imports to specific imports where that reduces bundle size.
6. Run the build and tests to confirm nothing broke.

Show bundle size before and after.
```

12
/ 31

Set timeouts for external API calls.

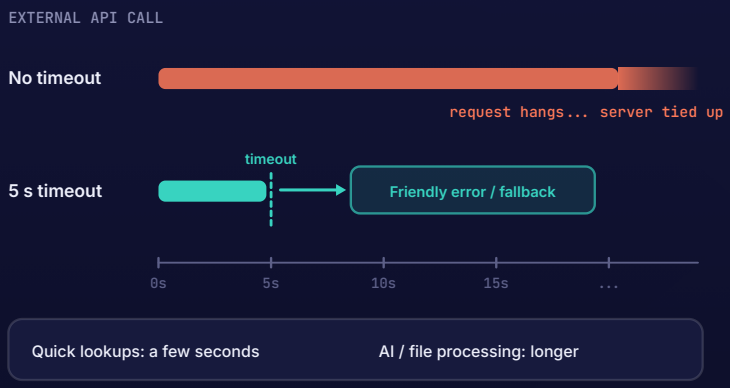
Backend IMPACT High EFFORT Low

WHAT IT IS

Give every call to a third-party service (payments, email, AI, maps) a maximum time to wait before giving up.

WHY IT MATTERS

Without a timeout, one slow external service can hang your requests and tie up your server, making the whole app feel broken.



A timeout turns a hung request into a quick fallback.

HOW TO DO IT

- Set a timeout on every outbound request (for example with `AbortSignal.timeout()`).
- Use short timeouts for quick lookups and longer ones for AI or file processing.
- Show a clear error or fallback when a call times out, and retry only when it's safe.

WATCH OUT

Many HTTP clients wait much longer than users will by default. Retrying non-idempotent calls (like payments) after a timeout can cause duplicates unless you use idempotency keys.

HOW TO VERIFY

Point a call at a deliberately slow endpoint: your app should give up at the timeout and show a friendly error.

COPY-PASTE PROMPT

```
Add timeouts to every external API call in my app.

1. Find every outbound HTTP request and third-party SDK call (payments, email, AI, storage, maps, webhooks, etc.).
2. Add an explicit timeout to each: a few seconds for quick lookups, longer for AI generation or file processing.
3. Route them through one shared helper so timeouts are consistent.
4. On timeout, return a clear error or fallback to the user instead of hanging.
5. Retry only safe, idempotent calls, with backoff and a max attempt count. Never blindly retry payments.

List each call with the timeout you chose and why.
```

13
/ 31

Stream AI responses token by token.

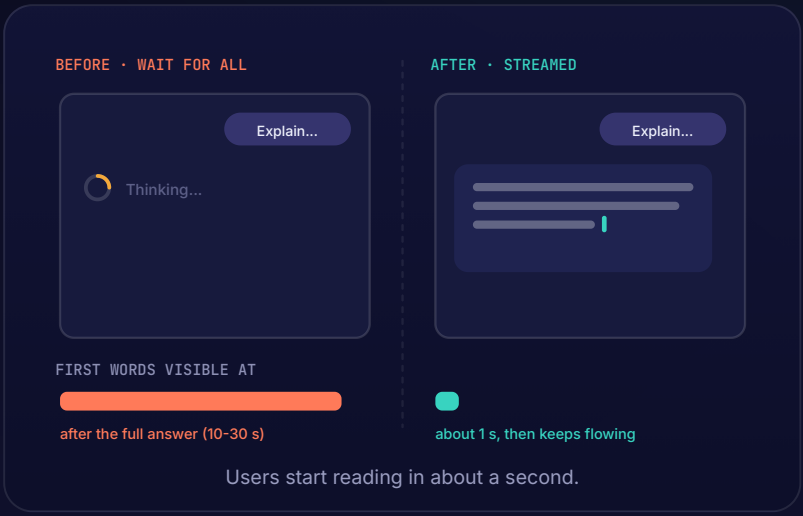
AI · IMPACT High · EFFORT Medium

WHAT IT IS

Show AI output word by word as it's generated, instead of waiting for the full answer and showing it all at once.

WHY IT MATTERS

The first words appear in about a second instead of after 10–30 seconds, so the app feels dramatically faster even though the total time is similar.



HOW TO DO IT

- Enable streaming in your AI provider's API call.
- Send the stream to the browser (for example with server-sent events) and render text as it arrives.
- Stop generation when the user cancels or leaves the page.

WATCH OUT

Some proxies, compression settings, or hosting setups buffer responses, which breaks streaming. Check that your host supports streaming responses.

HOW TO VERIFY

Ask a long question: text should start appearing within a second or two and keep flowing.

COPY-PASTE PROMPT

```
Make all AI responses in my app stream token by token.

1. Find every place where my app calls an AI model and waits for the full response.
2. Enable streaming in the API call and forward the stream to the frontend (server-sent
events or a streamed response).
3. Render text as it arrives, including markdown, without the layout jumping around.
4. Add a Stop button and abort the upstream AI request when the user cancels or navigates
away.
5. Make sure nothing buffers the stream (proxies, compression, or my host's settings).
6. Handle errors mid-stream gracefully.

Keep my existing prompts and model settings unchanged.
```

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/ 31

Update the UI before the server responds.

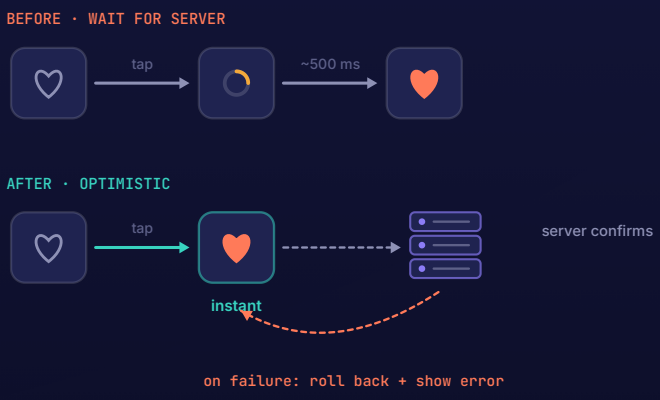
Frontend IMPACT Medium EFFORT Medium

WHAT IT IS

Called an optimistic update: show the result of an action (like, add to cart, rename) instantly, then confirm it with the server in the background.

WHY IT MATTERS

Actions feel instant instead of waiting for a round trip to the server.



The UI updates first; the server confirms after.

HOW TO DO IT

- Update local state right away when the user acts.
- Send the request in the background.
- If it fails, roll back to the previous state and show an error.

WATCH OUT

This is a display trick only. The server must still validate everything. Don't use it for payments or actions where showing a false success would be harmful.

HOW TO VERIFY

Throttle the network and click the action: the UI should update immediately and roll back cleanly if you force a failure.

COPY-PASTE PROMPT

```
Add optimistic UI updates to my app's quick actions.

1. Find small, low-risk actions where users currently wait for the server: likes, toggles, adding/removing items, renaming, reordering, marking as done.
2. For each, update the UI immediately, send the request in the background, and roll back with an error message if it fails.
3. Keep the server as the source of truth: refresh or reconcile with the real response when it arrives.
4. Do NOT apply this to payments, deletions that can't be undone, or anything security-sensitive.

List each action you made optimistic.
```

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/ 31

Lazy-load images below the screen.

Assets IMPACT Medium EFFORT Low

WHAT IT IS

Only load images and videos when the user scrolls near them, instead of downloading everything when the page opens.

WHY IT MATTERS

The first screen loads faster because the browser isn't busy downloading images nobody has seen yet.



Only what's on screen loads up front.

HOW TO DO IT

- Add `loading="lazy"` to images and iframes that start below the first screen.
- For videos, use `preload="none"` (or `metadata`) with a poster image.
- Always set width and height so the page doesn't jump when images load.

WATCH OUT

Never lazy-load the main image at the top of the page (usually the Largest Contentful Paint element). It makes it appear later. Give it high fetch priority instead.

HOW TO VERIFY

Open the Network tab and reload: below-the-fold images should only start downloading as you scroll.

COPY-PASTE PROMPT

```
Lazy-load images, videos, and embeds that are below the first screen.

1. Find all images, videos, and iframes (embeds, maps, video players) across my pages.
2. Add loading="lazy" to images and iframes that start below the first screen.
3. For videos, use preload="none" or "metadata" with a poster image.
4. Do NOT lazy-load the main image at the top of each page; give it fetchpriority="high" instead.
5. Make sure every image has width/height or an aspect ratio to prevent layout shift.

List which images you lazy-loaded and which you prioritized.
```

16 / 31

Run independent requests in parallel.

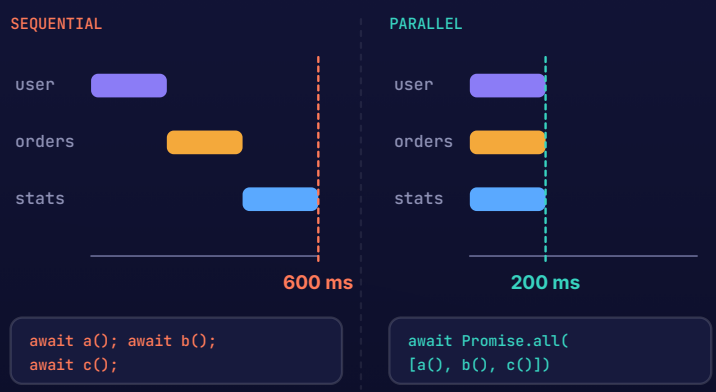
Backend IMPACT High EFFORT Low

WHAT IT IS

When several requests or queries don't depend on each other, start them all at once instead of waiting for each to finish before starting the next.

WHY IT MATTERS

Three 200 ms calls take about 600 ms one after another, but about 200 ms in parallel.



Independent calls run together, not one after another.

HOW TO DO IT

- Replace chains of `await` with `Promise.all` (or `Promise.allSettled` if some may fail).
- On the frontend, start fetches at the page level instead of nesting them in child components (request waterfalls).
- Cap concurrency when there are many calls, so you don't overload a service.

WATCH OUT

Only parallelize work that is truly independent. If request B needs the result of request A, it must wait.

HOW TO VERIFY

Look at the Network waterfall or your traces: independent requests should start at the same time.

COPY-PASTE PROMPT

```
Find sequential requests that could run in parallel.

1. Find code where independent awaits run one after another (API calls, database queries, file reads),
in both frontend and backend.
2. Run truly independent ones together with Promise.all, or Promise.allSettled when partial failure is
acceptable.
3. Find frontend request waterfalls (a child component only fetching after its parent finishes) and
start those fetches earlier/together.
4. If there are many calls (e.g., in a loop), limit concurrency so I don't overload any service.
5. Don't change anything where one call depends on another's result.

Show each change with an estimate of the time saved.
```

Halfway there. Enjoying the toolkit?

Follow @breakingbreach for more guides on building fast, secure, and cost-efficient apps.

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Add database indexes for common queries.

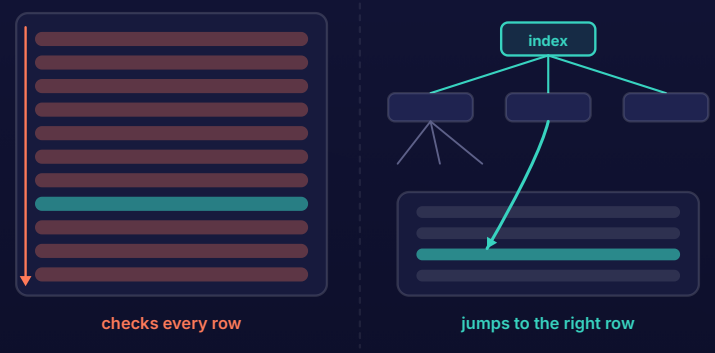
Database IMPACT High EFFORT Low

WHAT IT IS

An index lets the database find matching rows directly, instead of scanning the entire table for every query.

WHY IT MATTERS

A missing index can make a query hundreds of times slower once a table grows to thousands or millions of rows.



An index finds rows directly instead of scanning.

HOW TO DO IT

- Index columns used in `WHERE`, `JOIN`, and `ORDER BY`, especially foreign keys.
- Use composite indexes for queries that filter on several columns together (column order matters).
- Check slow queries with `EXPLAIN` to confirm the index is used.

WATCH OUT

Every index slows down writes and uses storage, so don't index everything. On large production tables, build indexes without locking the table (for example, `CREATE INDEX CONCURRENTLY` in Postgres).

HOW TO VERIFY

Run `EXPLAIN ANALYZE` on a slow query before and after: it should switch from a sequential scan to an index scan.

COPY-PASTE PROMPT

```
Add the right database indexes for my app's queries.

1. Read my schema and all the queries in my code (including ORM queries).
2. Find columns used in WHERE, JOIN, and ORDER BY that have no index, especially foreign keys and columns used for lookups (email, slug, user_id, created_at).
3. Propose indexes, including composite indexes in the right column order, and explain why each is needed.
4. Don't add duplicate or unnecessary indexes; every index slows down writes.
5. Create them as a migration, and use a non-locking method on large production tables (e.g., CREATE INDEX CONCURRENTLY in Postgres).

Show me EXPLAIN output before and after for the slowest queries.
```

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/ 31

Trace slow backend requests with OpenTelemetry.

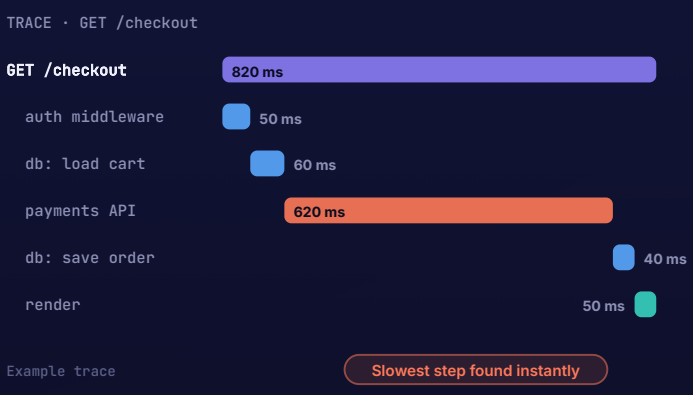
Monitoring IMPACT High EFFORT Medium

WHAT IT IS

Tracing records how long each step of a request takes (database queries, external calls, your own code), so you can see exactly where time goes. OpenTelemetry is an open standard for collecting traces.

WHY IT MATTERS

Instead of guessing why an endpoint is slow, you see a timeline showing the exact query or call that's responsible.



A trace shows exactly which step is slow.

HOW TO DO IT

- Add OpenTelemetry instrumentation for your HTTP server, database client, and outbound HTTP calls.
- Send traces to a tracing backend of your choice.
- Add custom spans around important business logic.

WATCH OUT

Sample traces in production to control volume and cost, and never put passwords, tokens, or personal data in trace attributes.

HOW TO VERIFY

Make a few requests and open your tracing backend: you should see each request broken into timed steps.

COPY-PASTE PROMPT

```
Add OpenTelemetry tracing to my backend.

1. Set up the OpenTelemetry SDK for my language and framework, with automatic instrumentation for incoming HTTP requests, database queries, and outbound HTTP calls.
2. Add custom spans around important logic (e.g., checkout, AI calls, file processing).
3. Configure an exporter, explain what I need to configure for a tracing backend, and read all settings from environment variables.
4. Use sampling in production so trace volume and cost stay reasonable.
5. Make sure no secrets, tokens, passwords, or personal data end up in span attributes or logs.

When done, tell me how to find my 5 slowest endpoints.
```

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/ 31

Place servers close to the database.

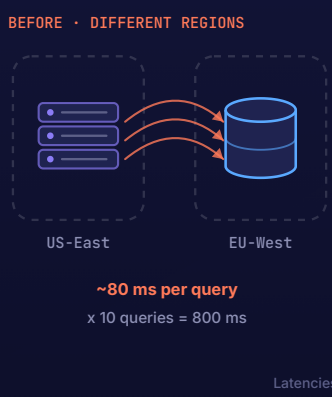
Infra IMPACT High EFFORT Low

WHAT IT IS

Run your app server (or serverless functions) in the same region as your database.

WHY IT MATTERS

Every query is a round trip. If your server is in one region and your database in another, a page with 10 queries can add hundreds of milliseconds.



Every query pays the distance between server and database.

HOW TO DO IT

- Check which region your database is in and which region your server or functions run in.
- Move one of them so they match.
- If users are worldwide, use a CDN for static content and read replicas for reads in other regions.

WATCH OUT

Running database-heavy code at the edge, far from the database, often makes things slower, not faster.

HOW TO VERIFY

Compare average query time or total server response time before and after moving.

COPY-PASTE PROMPT

```
Check whether my app servers and database are in the same region.

1. Look at my hosting, serverless, and database config files and tell me which region each part of my app runs in.
2. If the server/functions and database are in different regions, explain the latency cost and how to move them together.
3. Find any edge functions or middleware that make database calls, and suggest running those in the database's region instead.
4. If my users are global, suggest whether read replicas or caching would help.

Give me exact steps for any changes I need to make in my hosting or database dashboard.
```

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/ 31

Render pages on the server or at build time.

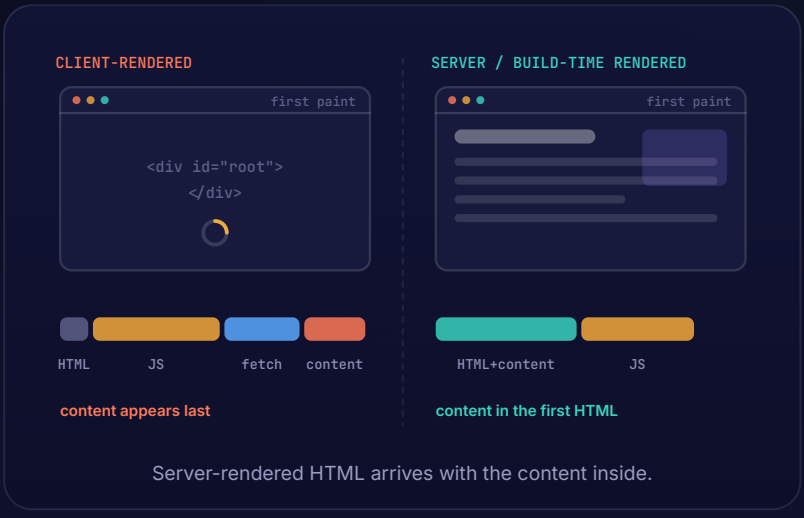
Frontend IMPACT High EFFORT Medium

WHAT IT IS

Send ready-made HTML instead of an empty page that the browser has to build with JavaScript. Pages that rarely change can be generated once at build time.

WHY IT MATTERS

Users see content much sooner, and search engines and link previews can read your pages.



Server-rendered HTML arrives with the content inside.

HOW TO DO IT

- Generate static pages (blog, docs, marketing pages) at build time.
- Render dynamic but public pages on the server.
- Keep client-side rendering for highly interactive, logged-in screens where SEO doesn't matter.

WATCH OUT

Be careful not to include secrets or other users' private data in server-rendered HTML or the page's embedded data.

HOW TO VERIFY

Use View Page Source: your main content should be in the HTML, not just an empty root element.

COPY-PASTE PROMPT

```
Improve how my pages are rendered.

1. List every route/page and how it's currently rendered (client-side, server-side, or
static).
2. Recommend the best mode for each: static generation for content that rarely changes, server
rendering for public dynamic pages, client rendering only for interactive logged-in screens.
3. Implement the changes using my framework's built-in features, adding revalidation for
static pages that change occasionally.
4. Make sure no secrets or private user data end up in the rendered HTML or embedded page
data.
5. Add proper titles and meta tags while you're there.

Show me the list of pages and the rendering mode you chose for each.
```

21
/ 31

Load third-party scripts after the page.

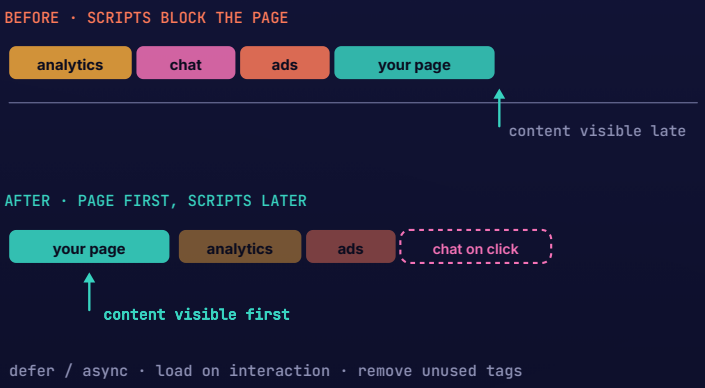
Frontend IMPACT High EFFORT Low

WHAT IT IS

Delay analytics, chat widgets, ads, and other third-party scripts until after your page has loaded.

WHY IT MATTERS

Third-party scripts are often the biggest cause of slow loads and laggy taps, and you don't control their size.



Your content loads before third-party scripts.

HOW TO DO IT

- Add `defer` or `async` to script tags, or use your framework's script component with a lazy loading strategy.
- Load non-essential widgets (chat, video embeds) only when the user interacts, using a lightweight placeholder until then.
- Remove any trackers or tools you no longer use.

WATCH OUT

`defer` keeps scripts in order and runs them after the HTML is parsed; `async` runs them as soon as they download, in any order. Pick the one that matches each script.

HOW TO VERIFY

Run a performance audit before and after: blocking time and main-thread work from third parties should drop.

COPY-PASTE PROMPT

```
Optimize third-party scripts in my app.

1. List every third-party script and embed (analytics, tag managers, chat, ads, video players, maps, social widgets, A/B testing).
2. Flag any that are unused and recommend removing them.
3. Load the rest with defer/async or my framework's lazy script strategy so they never block the first render.
4. Replace heavy embeds (chat widgets, video players, maps) with a lightweight placeholder that loads the real thing on click or when scrolled into view.
5. Keep anything required for consent or security working correctly.

Show the list with what you changed for each script.
```

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Split JavaScript by page.

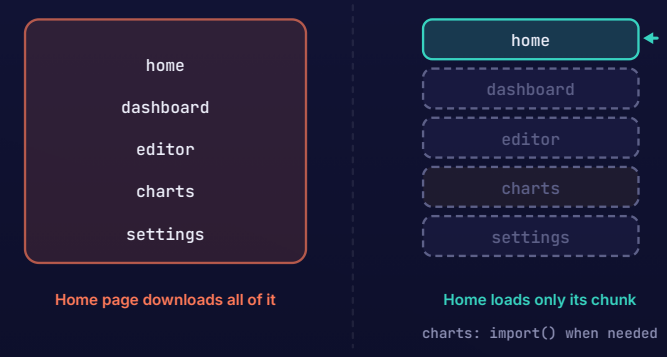
Frontend · IMPACT High · EFFORT Medium

WHAT IT IS

Also called code splitting: each page downloads only the JavaScript it needs, instead of one giant bundle for the whole app.

WHY IT MATTERS

Your home page shouldn't have to download the code for your dashboard, editor, and settings pages.



Each page downloads only the code it needs.

HOW TO DO IT

- Use route-based splitting (most modern frameworks do this automatically per page).
- Load heavy components (charts, rich text editors, maps, PDF viewers) with dynamic `import()` only when needed.
- Keep heavy libraries out of shared layouts that every page loads.

WATCH OUT

Splitting into too many tiny chunks can add extra requests. Focus on large, rarely used components.

HOW TO VERIFY

Use a bundle analyzer: each page's initial JavaScript should be smaller, and heavy libraries should sit in separate chunks.

COPY-PASTE PROMPT

```
Reduce how much JavaScript each page loads.

1. Analyze my production bundle and list the largest chunks and libraries, and which pages load them.
2. Make sure routes are code-split so each page only loads its own code.
3. Lazy-load heavy components (charts, editors, maps, PDF viewers, modals, admin-only screens) with dynamic import() and a loading placeholder.
4. Move heavy imports out of shared layouts and root components.
5. Don't split tiny components; focus on the biggest wins.

Show the initial JS size per page before and after.
```

23 / 31

Virtualize long lists.

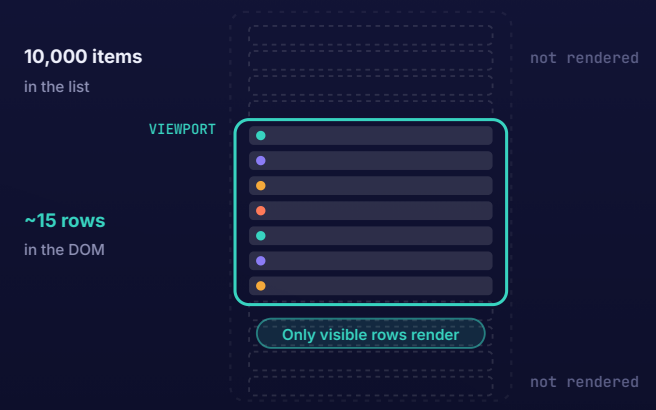
Frontend · IMPACT Medium · EFFORT Medium

WHAT IT IS

Only render the rows that are visible on screen (plus a few extra), even when a list or table has thousands of items.

WHY IT MATTERS

Rendering thousands of rows at once makes scrolling laggy and can freeze the page on slower phones.



Thousands of items, only a handful of rows in the DOM.

HOW TO DO IT

- Use a list-virtualization library or your UI kit's virtualized list/table component.
- Apply it to lists with hundreds of items or more.
- Combine it with pagination or infinite loading so you don't fetch everything either.

WATCH OUT

Browser find-in-page (Ctrl/Cmd+F) can't find rows that aren't rendered, and variable-height rows need extra care. Keep keyboard navigation and accessibility working.

HOW TO VERIFY

Inspect the DOM while scrolling: only a small, roughly constant number of rows should exist at any time.

COPY-PASTE PROMPT

```
Virtualize long lists and tables in my app.

1. Find every list, table, grid, feed, or dropdown that can contain hundreds of items or more.
2. Implement virtualization so only visible rows (plus a small overscan) are rendered, using a well-maintained library that fits my stack.
3. Handle variable row heights correctly if my rows aren't all the same size.
4. Keep scroll position, keyboard navigation, selection, and accessibility working.
5. Combine with pagination or infinite loading where data comes from the server.

List every list you virtualized.
```

24 / 31

Select only the columns you need.

Database · IMPACT Medium · EFFORT Low

WHAT IT IS

Query only the fields a page actually uses, instead of `SELECT *` or fetching whole objects.

WHY IT MATTERS

Less data is read from disk, sent over the network, and serialized, especially when tables have large text or JSON columns.



Fetch the fields you show, nothing more.

HOW TO DO IT

- Replace `SELECT *` with explicit column lists.
- Use your ORM's select option to pick fields.
- Avoid loading large columns (body text, JSON blobs, files) on list pages.

WATCH OUT

This also improves security: it stops fields like password hashes or internal notes from accidentally reaching the frontend.

HOW TO VERIFY

Compare response size and query time for your main list endpoints before and after.

COPY-PASTE PROMPT

```
Make my database queries fetch only the columns they need.

1. Find every SELECT * and every ORM query that loads full records.
2. For each, check which fields the calling code and API response actually use, and
select only those.
3. Pay special attention to large columns (long text, JSON, binary data) on list pages.
4. Make sure sensitive fields (password hashes, tokens, internal notes) are never
selected for responses that go to the client.
5. Keep API responses exactly the same for anything the frontend uses.

List each query you changed and the fields removed.
```

25
/ 31

Use database connection pooling.

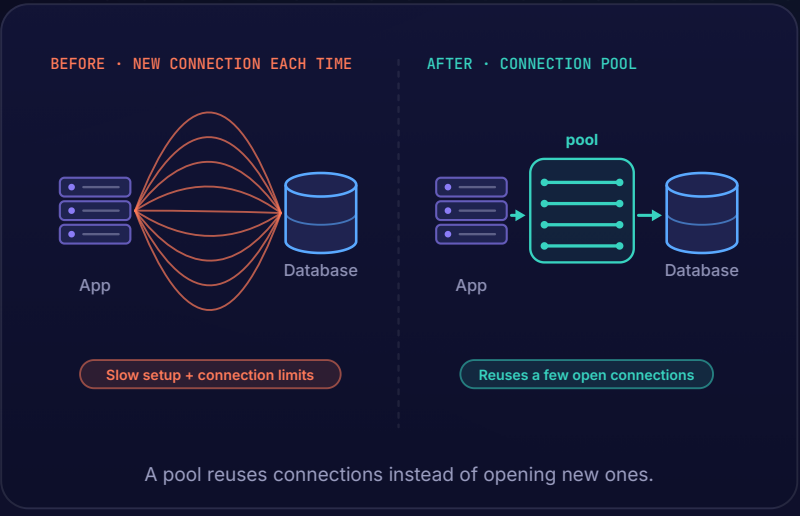
Database · IMPACT High · EFFORT Low

WHAT IT IS

Reuse a small set of open database connections instead of opening a brand-new connection for every request.

WHY IT MATTERS

Opening a connection takes time, and too many connections can crash your database or hit its connection limit.



A pool reuses connections instead of opening new ones.

HOW TO DO IT

- Create one shared database client per server process instead of one per request.
- On serverless platforms, use a connection pooler (many database providers offer a pooled connection URL).
- Set a sensible pool size that stays under your database's connection limit.

WATCH OUT

Some poolers run in transaction mode, which may not support prepared statements or session features. Check your database provider's docs for the right settings.

HOW TO VERIFY

Watch your database's active connection count under load: it should stay stable instead of climbing.

COPY-PASTE PROMPT

```
Set up proper database connection pooling.

1. Find where my app creates database connections or clients and check whether a new one is created per request or per function call.
2. Refactor to a single shared client/pool per process, reused across requests (including during development hot reloads).
3. If I'm on serverless, configure a connection pooler or my provider's pooled connection URL, and note any settings it requires (e.g., for prepared statements).
4. Set a pool size that stays safely under my database's connection limit.
5. Read connection settings from environment variables.

Explain how to monitor active connections.
```

26
/ 31

Prevent unnecessary re-renders.

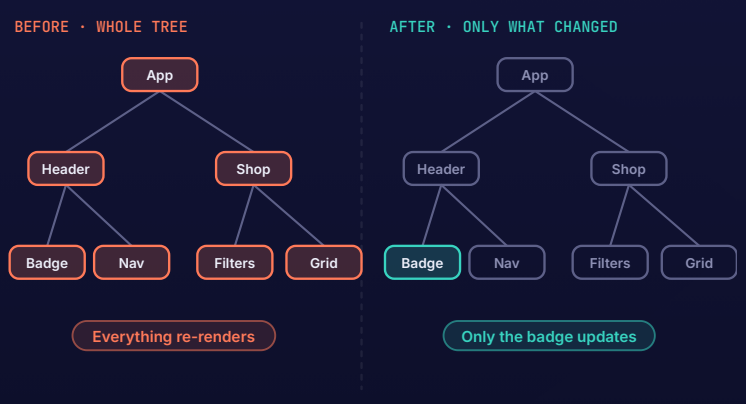
Frontend IMPACT Medium EFFORT Medium

WHAT IT IS

Stop components from re-rendering when nothing they display has changed.

WHY IT MATTERS

Unnecessary re-renders make typing, scrolling, and clicking feel laggy, especially on slower phones.



Only components whose data changed re-render.

HOW TO DO IT

- Use your framework's profiler to find components that re-render too often.
- Keep state close to where it's used, and split large contexts that change often.
- Memoize genuinely expensive components and calculations, and keep callback and object references stable.

WATCH OUT

Don't memoize everything. It adds complexity and can be slower. Fix the components the profiler flags first.

HOW TO VERIFY

Record an interaction in the profiler: only the components that actually changed should re-render.

COPY-PASTE PROMPT

```
Find and fix unnecessary re-renders in my frontend.

1. Look for common causes: state stored too high up, large contexts that change often, new objects/arrays/functions created on every render, expensive calculations repeated each render, and missing or unstable list keys.
2. Fix them by moving state closer to where it's used, splitting contexts, memoizing expensive components and calculations, and keeping references stable.
3. Only memoize where it actually helps; don't wrap everything.
4. Keep behavior identical.

Tell me how to confirm the improvement with my framework's profiler, and list the components you changed.
```

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Enable browser caching for static files.

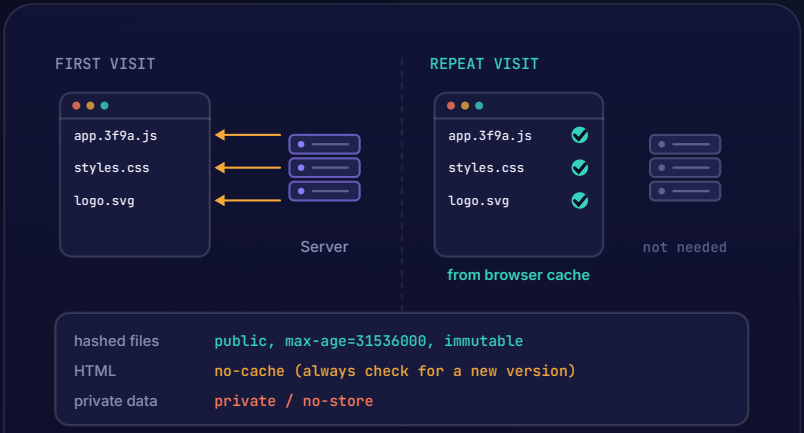
Network · IMPACT Medium · EFFORT Low

WHAT IT IS

Tell browsers to keep copies of static files, so returning visitors don't download them again.

WHY IT MATTERS

Repeat visits load almost instantly because most files come straight from the browser's cache.



Returning visitors get static files from their cache.

HOW TO DO IT

- For versioned files with a hash in the name (like `app.3f9a.js`), use `Cache-Control: public, max-age=31536000, immutable`.
- For HTML, use `no-cache` so the browser always checks for a new version.
- For private or logged-in responses, use `private` or `no-store`.

WATCH OUT

Only long-cache files whose names change when their content changes. Otherwise users may be stuck with old versions of your app.

HOW TO VERIFY

Reload the page and check the Network tab: static files should show as served from memory or disk cache.

COPY-PASTE PROMPT

```
Set up correct browser caching headers for my app.

1. Check the current Cache-Control headers for my HTML, JS, CSS, images, fonts, and API responses.
2. For versioned/hashed static files, set: Cache-Control: public, max-age=31536000, immutable.
3. For HTML pages, set Cache-Control: no-cache so users always get the latest version.
4. For API responses and anything personalized or behind login, use private or no-store.
5. Make sure no unversioned file gets a long cache time.

Show me the final header for each file type and where it's configured.
```

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/ 31

Load-balance traffic across multiple app instances.

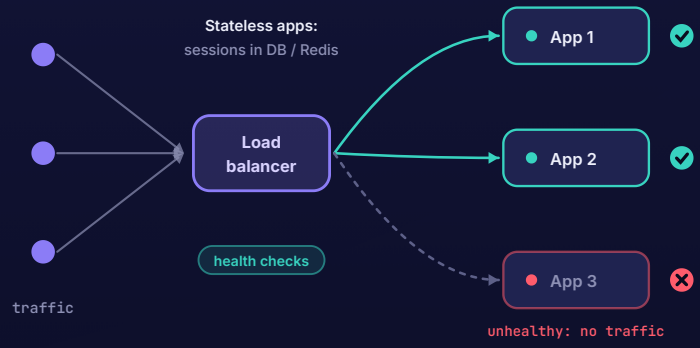
Infra · IMPACT Medium · EFFORT Medium

WHAT IT IS

Run more than one copy of your app and let a load balancer spread incoming requests between them.

WHY IT MATTERS

No single server gets overwhelmed during traffic spikes, and if one instance crashes the others keep your app online.



Traffic is spread across healthy instances only.

HOW TO DO IT

- Make your app stateless: store sessions, uploads, and caches in a database, object storage, or Redis, not in server memory or local disk.
- Add a health-check endpoint so the load balancer only sends traffic to healthy instances.
- On most hosting platforms, this just means increasing the instance count or enabling autoscaling.

WATCH OUT

Most early-stage apps don't need this yet. If you enable autoscaling, set a maximum instance count so a traffic spike can't run up your bill.

HOW TO VERIFY

Run two or more instances, stop one, and confirm the app keeps working without errors or lost sessions.

COPY-PASTE PROMPT

```
Prepare my app to run on multiple instances behind a load balancer.

1. Find anything that stores state in server memory or on local disk (sessions, uploads, in-memory caches, rate-limit counters, websocket state, scheduled jobs) and move it to shared storage (database, object storage, or Redis).
2. Add a /health endpoint that checks the app and its database connection.
3. Make sure scheduled/cron jobs won't run multiple times when several instances are running.
4. Explain how to enable multiple instances or autoscaling on my host, and recommend a minimum and a maximum instance count to control costs.

List every stateful piece you found and how you fixed it.
```

29
/ 31

Self-host fonts and limit font weights.

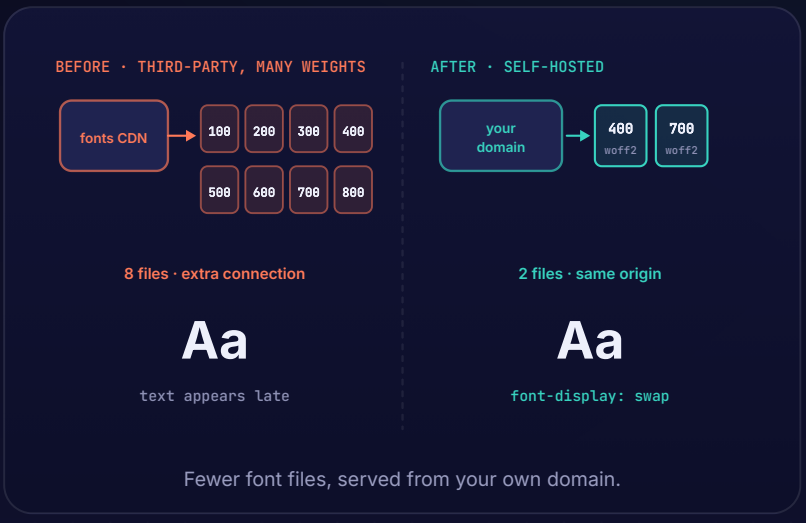
Assets IMPACT Medium EFFORT Low

WHAT IT IS

Serve font files from your own domain and only load the weights and styles you actually use.

WHY IT MATTERS

Fonts can block text from appearing. Fewer, smaller font files mean text shows up faster, with no extra connection to a third-party server.



Fewer font files, served from your own domain.

HOW TO DO IT

- Download the fonts in WOFF2 format and serve them yourself, or use your framework's font helper, which often self-hosts automatically.
- Limit to 2–3 weights, or use one variable font file.
- Use `font-display: swap`, and preload only the main font file.

WATCH OUT

Check the font's license allows self-hosting. Preloading too many fonts slows the page down instead of speeding it up.

HOW TO VERIFY

In the Network tab, fonts should load from your own domain, and only a few font files should download.

COPY-PASTE PROMPT

```
Optimize the fonts in my app.

1. Find every font my app loads and every weight/style actually used in my CSS and components.
2. Self-host them as WOFF2 (or use my framework's built-in font optimization), removing requests to third-party font servers.
3. Remove unused weights and styles, keeping it to 2-3 weights or a single variable font.
4. Use font-display: swap and preload only the primary font file.
5. Subset to the character sets I actually need, if supported.
6. Add a matching fallback font stack to minimize layout shift when the font loads.

Show which font files are loaded before and after.
```

30 / 31

Track real-user speed with a performance monitoring tool.

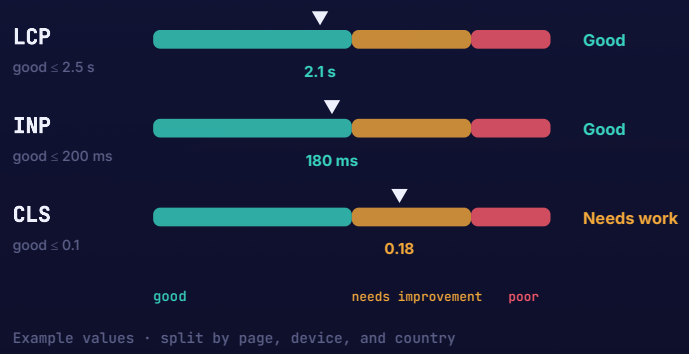
Monitoring · IMPACT High · EFFORT Low

WHAT IT IS

Measure how fast your app actually is for real users on their real devices and networks (called Real User Monitoring), not just on your fast laptop.

WHY IT MATTERS

Your users' phones and connections are usually much slower than yours. Real data shows which pages, devices, and countries need work.



Track Core Web Vitals from real visits.

HOW TO DO IT

- Track Core Web Vitals: LCP (loading), INP (responsiveness), and CLS (visual stability).
- Aim for good scores at the 75th percentile: LCP ≤ 2.5 s, INP ≤ 200 ms, CLS ≤ 0.1.
- Break results down by page, device type, and country.

WATCH OUT

Respect privacy laws and consent requirements, and don't collect personal data in performance events.

HOW TO VERIFY

After a few days of traffic, you should see Core Web Vitals per page in your monitoring dashboard.

COPY-PASTE PROMPT

```
Set up real-user performance monitoring for my app.

1. Recommend a performance monitoring approach that fits my stack and budget.
2. Collect Core Web Vitals (LCP, INP, CLS) plus page load time from real users, tagged by page/route, device type, and connection type.
3. Add backend response time tracking for my API routes if possible.
4. Respect user consent and don't collect personal data in performance events.
5. Set targets using Google's "good" thresholds at the 75th percentile: LCP ≤ 2.5 s, INP ≤ 200 ms, CLS ≤ 0.1.

Explain how to read the dashboard and find my worst pages.
```

31
/ 31

Test on slow phones and networks.

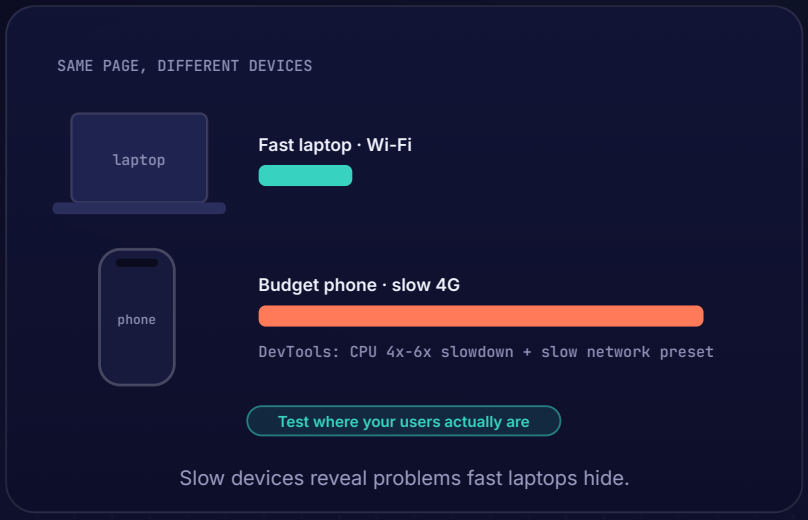
Monitoring IMPACT High EFFORT Low

WHAT IT IS

Test your app the way most of the world actually uses it: on mid-range phones over slow mobile connections.

WHY IT MATTERS

An app that feels instant on a new laptop over fast Wi-Fi can feel painfully slow on a budget phone over a weak connection.



Slow devices reveal problems fast laptops hide.

HOW TO DO IT

- Use your browser DevTools to throttle CPU (4x–6x slowdown) and network (slow 4G/3G presets).
- Run a mobile performance audit on your key pages.
- Test on at least one real mid-range or budget Android phone.

WATCH OUT

Emulation is only an approximation. Real devices often reveal problems DevTools doesn't.

HOW TO VERIFY

Your key pages should load and respond acceptably with throttling on, and you can set performance budgets to keep them that way.

COPY-PASTE PROMPT

```
Help me test and improve my app on slow phones and networks.

1. Give me a step-by-step test plan using browser DevTools: CPU throttling (4x–6x) and slow network presets, for my 5 most important pages and user flows.
2. Run (or tell me how to run) a mobile performance audit on those pages and summarize the biggest issues.
3. Rank the fixes by impact on slow devices: JS size, images, fonts, third-party scripts, long tasks, and request waterfalls.
4. Propose performance budgets (e.g., max JS per page, max image weight, LCP target) and, if possible, add a check that fails the build when a budget is exceeded.

Start with the fixes that help slow devices the most.
```

PROMPT 32

The final *check.*

When you've worked through the list, run this prompt to measure your results, confirm nothing broke, and catch any security mistakes the changes may have introduced.

FINAL CHECK PROMPT

```
We've finished the performance work. Now verify everything.

1. Re-run the same measurements from PERFORMANCE.md and fill in the "After" column.
2. Run the full test suite and build, and click through the main user flows to confirm nothing broke.
3. Double-check for security mistakes introduced by the changes: caches keyed per user, no private data
cached by the CDN or browser, no secrets in server-rendered HTML, and no unsafe retries on payments.
4. Summarize the before/after results and anything still worth improving.
```

Targets to aim for

METRIC	WHAT IT MEASURES	GOOD
LCP	Largest Contentful Paint: how fast the main content appears	≤ 2.5 s
INP	Interaction to Next Paint: how fast the page responds to taps and clicks	≤ 200 ms
CLS	Cumulative Layout Shift: how much the page jumps around	≤ 0.1
TTFB	Time to First Byte: how fast the server starts responding	≤ 0.8 s
Lighthouse	Lab performance score from a mobile audit	90+

Web Vitals thresholds are Google's "good" ratings, measured at the 75th percentile of real page loads.

> **Speed and safety go together**
> Several fixes in this toolkit (caching, CDNs, server rendering, retries) can leak data or cause duplicate charges if they're done carelessly. Each card's "Watch out" box flags those risks. Read them before you ship.

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